

ia-26/functions.py

```

1 import numpy as np
2
3
4 # =====
5 # Benchmark Test Functions
6 # =====
7
8
9 def f1(x):
10     """
11     Sphere Function
12     """
13     x = np.asarray(x)
14
15     return np.sum(x ** 2)
16
17
18 def f2(x):
19     """
20     Schaffer Function N.2
21
22     Agora recebe vetor:
23     x = [x1, x2]
24     """
25
26     x = np.asarray(x)
27
28     x1 = x[0]
29     x2 = x[1]
30
31     numerator = (
32         np.sin(np.sqrt(x1**2 + x2**2))**2
33         - 0.5
34     )
35
36     denominator = (
37         1 + 0.001 * (x1**2 + x2**2)
38     )**2
39
40     return 0.5 + numerator / denominator
41
42
43 def f3(x):
44     """
45     Ackley Function
46     """
47
48     x = np.asarray(x)
49
50     n = len(x)

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51
52     term1 = (
53         -20
54         * np.exp(
55             -0.2 * np.sqrt(np.sum(x**2) / n)
56         )
57     )
58
59     term2 = (
60         -np.exp(
61             np.sum(
62                 np.cos(2 * np.pi * x)
63             ) / n
64         )
65     )
66
67     return term1 + term2 + 20 + np.e
68
69
70 def f4(x):
71     """
72     Rosenbrock Function
73     """
74
75     x = np.asarray(x)
76
77     return np.sum(
78         100 * (x[:-1]**2 - x[1:])**2
79         + (x[:-1] - 1)**2
80     )
81
82
83 def f5(x):
84     """
85     Rastrigin Function
86     """
87
88     x = np.asarray(x)
89
90     return np.sum(
91         x**2
92         - 10 * np.cos(2 * np.pi * x)
93         + 10
94     )
95
96
97 def f6(x):
98     """
99     Griewank Function
100     """
101
102     x = np.asarray(x)
103

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104     n = len(x)
105
106     sum_term = np.sum(x**2) / 4000
107
108     prod_term = np.prod(
109         np.cos(
110             x / np.sqrt(np.arange(1, n + 1))
111         )
112     )
113
114     return sum_term - prod_term + 1
115
116 def u(x, a, k, m):
117
118     if x > a:
119         return k * (x - a) ** m
120
121     elif x < -a:
122         return k * (-x - a) ** m
123
124     else:
125         return 0
126
127 def f7(x):
128     """
129     Penalized Function
130     """
131
132     x = np.asarray(x)
133
134     n = len(x)
135
136     y = 1 + (x + 1) / 4
137
138     first_term = (
139         10 * np.sin(np.pi * y[0])**2
140     )
141
142     middle_sum = np.sum(
143         (y[:-1] - 1)**2
144         * (
145             1
146             + 10
147             * np.sin(np.pi * y[1:])**2
148         )
149     )
150
151     last_term = (y[-1] - 1)**2
152
153     penalty = np.sum(
154         [
155

```

```

157         u(xi, 10, 100, 4)
158         for xi in x
159     ]
160 )
161
162 return (
163     (np.pi / n)
164     * (
165         first_term
166         + middle_sum
167         + last_term
168     )
169     + penalty
170 )
171
172 def f8(x):
173     """
174     Schwefel Function
175     """
176
177     x = np.asarray(x)
178
179     n = len(x)
180
181     return (
182         418.9829 * n
183         - np.sum(
184             x * np.sin(np.sqrt(np.abs(x)))
185         )
186     )
187

```